Aspa: An offline tool to generate, read and grade versioned and personalised multiple-choice exams

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Abstract

Grading paper-based multiple-choice exams requires either proprietary optical readers or tools that demand LaTeX or programming skills. Aspa is a local web application that covers the complete cycle: it generates exam versions from a spreadsheet question bank, prints answer sheets with fiducial markers and a QR code, reads scans and mobile-phone photographs with computer vision, grades with negative marking and partial credit, and supports the review of each sheet with the student. It also generates personalised exams that omit the questions each student has already passed during the course. All processing is local, so students' data never leave the teacher's computer.

Keywords: Multiple-choice exams, Optical mark recognition, Automated grading, Personalised assessment, Continuous assessment

Required Metadata

Current code version

1. Motivation and significance

Multiple-choice questions (MCQs) sample a broad syllabus in little time and are graded objectively, which explains their widespread use in higher education [1]. When exams are sat on paper, manual grading of large groups is slow and error-prone, and item analysis, which detects flawed questions and wrong keys [2], is seldom performed. Practices that improve fairness, such

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Nr.	Code metadata description	Please fill in this column
C1	Current code version	v1.0.0
C2	Permanent GitHub link to code/repository used for this code version	`https://github.com/lorenagmad/Aspa`
C3	Legal Code License	AGPL-3.0-or-later
C4	Code versioning system used	git
C5	Software code languages, tools, and services used	Python 3; Flask, OpenCV, NumPy, ReportLab, openpyxl, PyMuPDF; HTML, CSS, JavaScript
C6	Compilation requirements, operating environments & dependencies	Python $\geq$ 3.9 on Windows, macOS or Linux; dependencies in `requirements.txt`, installed by the launch scripts; a web browser
C7	If available Link to developer documentation/manual	`https://github.com/lorenagmad/Aspa#readme`
C8	Support email for questions	lorena.gutierrez@uca.es

Table 1: Code metadata (mandatory)

as distributing several versions of the exam or applying negative marking to discourage guessing [3], further increase the grading workload.

Continuous assessment adds a further requirement. Frequent low-stakes quizzes improve retention [4] and provide formative feedback [5, 6]. A consistent policy, in line with mastery learning [7], is to exempt each student from the final-exam questions whose content they have already demonstrated. On paper, this requires a different exam and answer key per student, which is impractical without software support.

Existing tools solve this problem only partially. Commercial optical mark recognition (OMR) products, such as Remark OMR or ZipGrade [8], are proprietary, and cloud services transfer personal data to third parties, which raises compliance issues under the GDPR [9]. Among open-source tools, Auto Multiple Choice [10] requires exams to be written in LaTeX and runs only on Linux and macOS; R/exams [11] requires R programming; FormScanner [12] and OMRChecker [13] read generic bubble sheets but neither generate nor grade versioned exams. To our knowledge, none supports exams personalised from the record of questions each student has passed.

Aspa was developed at the University of Cádiz under three requirements: (i) usable without programming, from the spreadsheets teachers already handle (question bank and class list); (ii) able to read sheets digitised by any means, from a sheet-fed scanner to a phone camera; and (iii) fully local.

The workflow consists of six steps: upload the bank and the class list; configure exam type, language, scoring and versions; generate the printable PDF; upload the scans, which are read and graded automatically, and review the flagged sheets; inspect item statistics and void questions or correct keys; and export the grades.

2. Software description

Aspa 1.0.0 is a Python application served by Flask on `127.0.0.1` and used through the browser. Launch scripts for Windows, macOS and Linux create a virtual environment and install the dependencies; a PyInstaller script builds a standalone executable. The interface is available in Spanish and English, and languages are added as JSON files.

2.1. Software architecture

The web layer (Figure 1) contains no exam logic, which is implemented in the `examtool` package. Two design decisions are central. First, the module `layout` is the single definition of the answer-sheet geometry, shared by the PDF generator (ReportLab), the reader (OpenCV [14]) and a scan simulator, so generator and reader cannot diverge. Second, each exam is a folder of plain files: settings, bank, class list, the versions frozen at generation time (seed, question and option order, keys), the readings of each page and the teacher's manual corrections. Grades are never stored; they are recomputed from these files. Hence, changing the scoring, voiding a question or correcting a key updates all grades immediately, and re-reading the scans never overwrites manual corrections.

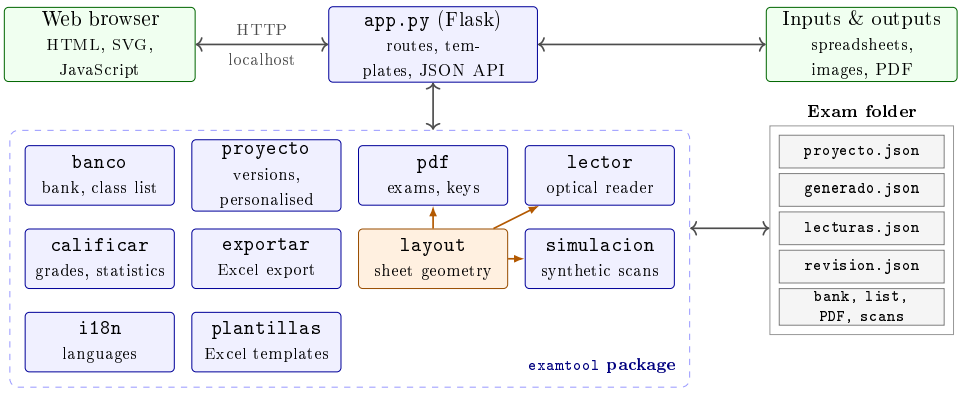


Figure 1: Architecture of Aspa. The sheet geometry (`layout`) is shared by generator, reader and simulator. Settings, frozen versions, readings and manual revisions are stored as JSON files; grades are recomputed from them.

2.2. Software functionalities

Question bank and class list. The bank is an Excel or CSV file with one row per question: identifier, topic, stem, up to four options, key (letters, or digits for legacy banks), group, fixed options and an optional explanation. Headers are matched by Spanish and English synonyms. Multiple-answer, true/false and fixed-order questions are supported; options such as "All/None of the above" are fixed automatically, and inconsistencies (duplicate identifiers, empty keys) are reported. The class list accepts the export of the university system or a simple sheet with each student's passed questions. Templates and help pages are built in.

Versions and personalised exams. Each version lists the question groups it draws from and is completed at random with ungrouped questions; before generation, Aspa reports the source of the questions of each version or the shortfall. Questions and options are shuffled with a recorded seed, and copies are interleaved (A, B, C...) and padded for duplex printing. A second PDF contains the keys and the mapping to bank identifiers. In personalised mode, each student is assigned to a version whose pool contains all their passed questions, balancing the load across versions; the exam includes that pool minus the passed questions, which count as correct, and prints the student's name. Non-personalised spare copies are also produced.

Answer sheet. Four ArUco markers [15] delimit the reading area; a QR code encodes exam, version, copy and, if personalised, the student's document; a ten-column digit grid records the identity document (national ID, residence card or passport digits); up to 30 questions are answered in boxes A–D, with a separate *Cancel* box per question and a clarification box (Figure 2). Instructions are printed in the exam language, independent of the interface language.

Optical mark recognition. Images and multi-page PDF files (rasterised at 220 dpi) are accepted. For each page, the reader: (1) detects the markers, with a contrast-enhanced (CLAHE) retry, and estimates the page homography with RANSAC from at least three markers, correcting rotation, scale and perspective; (2) rectifies the page at 8 px/mm and normalises illumination by dividing by a background estimate (morphological closing with a 7 mm kernel); (3) decodes the QR code; (4) fits each block of boxes to the printed boxes found by contour analysis (median shift, RANSAC affine transform and per-box snapping), which compensates paper curvature and lens distortion that a homography cannot model; and (5) classifies each box by its ink fraction with sheet-adaptive thresholds, marked above $\max(0.11, m + 0.10)$ and doubtful above $\max(0.04, m + 0.04)$, where m is

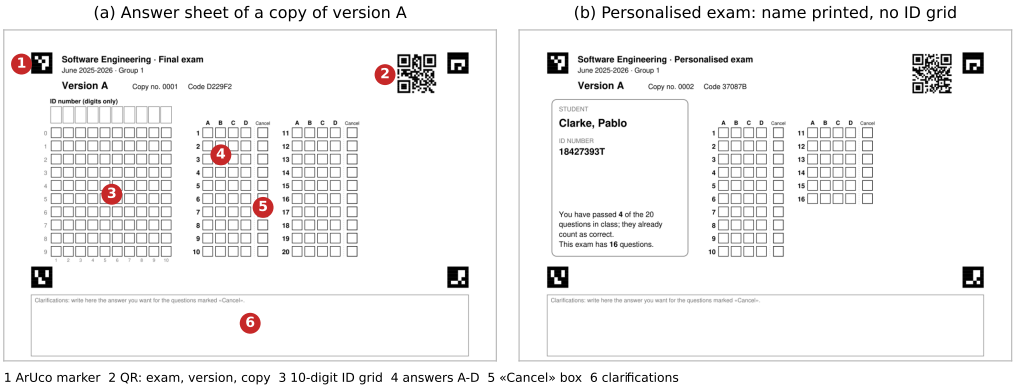


Figure 2: Answer sheets of (a) version A and (b) a personalised exam, in which the student's data replace the ID grid and the four questions passed in class are omitted.

the median fraction of the printed boxes. Blank pages and pages without markers are ignored.

Identification and incidences. Identity digits are matched with the class list. Columns with several marks act as wildcards and a unique one-digit mismatch is proposed; both cases are flagged. Each sheet is classified as correct, to review or unreadable, with the reasons: QR code missing or from another exam, unknown or unmarked document, doubtful marks, pending *Cancel* boxes or duplicate student.

Grading. With b points per correct answer, $w < 0$ per wrong answer, key C and marks M, a question scores 0 if $M = \varnothing$, b if $M = C$ and w otherwise when $|C| = 1$. For multiple-answer questions in partial-credit mode, with $h = |M \cap C|$ and $e = |M \setminus C|$, the score is w if $h = 0$ and $b \max(0, h - e)/|C|$ otherwise; in exact mode any $M \neq C$ scores w (Table 2). A marked *Cancel* box scores zero until the teacher enters the intended answer. The grade is scaled to the maximum attainable on non-voided questions, passed questions add b, and negative grades are optional.

Table 2: Score of a question with key AB ($b = 1$, $w = -0.25$) for different marks.

Marked	AB	A	B	ABC	AC	ABCD	C	CD	(blank)
Partial credit	1	0.5	0.5	0.5	0	0	−0.25	−0.25	0
Exact	1	−0.25	−0.25	−0.25	−0.25	−0.25	−0.25	−0.25	0

Review, statistics and export. The review screen (Figure 3) overlays the result on the rectified scan. Clicking a box or editing the table updates the grade; the teacher can reassign the student or the version and mark the sheet as reviewed. Hovering over a question displays its text and, if it was not answered correctly, the explanation from the bank. A student mode hides the editing controls. The statistics screen lists, for each bank question, success rate, mean score and option distribution in bank order, sorted by success rate; questions can be voided or rekeyed, and Aspa maps corrected keys onto the order of each version. Grades, class list with absences, answers, points per question, statistics and settings are exported to Excel.

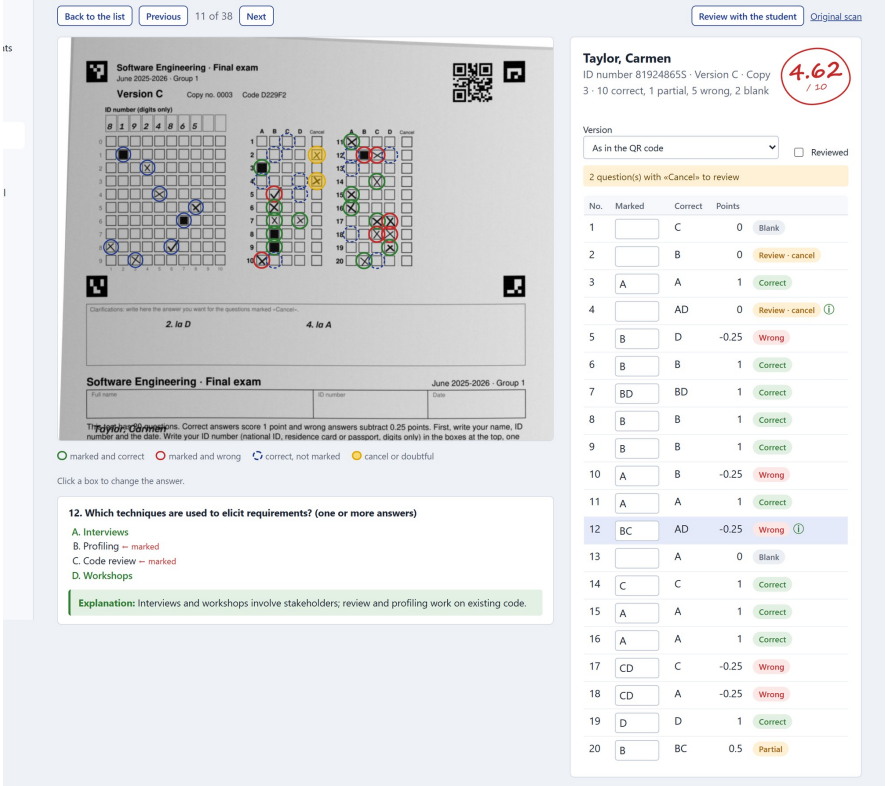


Figure 3: Review screen: marked and correct (green), marked and wrong (red), correct but unmarked (dashed blue) and *Cancel* (yellow). The selected question shows the explanation of the correct answer.

3. Illustrative examples

The examples use a 40-question bank on software engineering (groups T and R and 20 ungrouped questions) and 40 fictitious students. Data and scripts are available in `SoftwareX/experiments` for reproducibility.

Complete exam. Four versions of 20 questions were defined: A (group T), B (group R), C (T and R) and D (whole bank). Generating 40 copies (160 duplex pages) and the keys took 3.3 s. Thirty-six copies were filled in and degraded by the simulator and uploaded as 30 JPEG images and one PDF that also contained a blank page and a question page. The 38 pages were read in 11.5 s [processor model]: the two extra pages were ignored, the 720 answers were read without error and all students were identified. The 21 sheets flagged for review all contained *Cancel* marks introduced by the simulator. Voiding the question with the lowest success rate changed 18 grades without losing manual corrections, and corrected keys were mapped correctly onto each version. In personalised mode, Aspa produced 40 exams of 14 to 20 questions and three spare copies, identified every sheet by its QR code and counted the passed questions as correct. These steps belong to an end-to-end test suite of 67 checks run through the web interface, all of which passed.

Reading accuracy. Sheets were filled with known answers (crosses, filled boxes and ticks; blank, single and double answers; 5% *Cancel* marks) and read with the same function Aspa applies to real scans. First, 500 versioned and 125 personalised sheets were degraded with a random mixture of rotation (one in ten upside down), perspective, warping of up to 2 mm, shadow, grey paper, noise, blur, JPEG compression and ink from black pen to light pencil. Second, each defect was varied separately on 30 sheets per level. Table 3 summarises the results. In the mixed experiment, the only misread answer out of 12,066 was flagged as doubtful, and the unread sheets had part of the page outside the image, a condition that Aspa reports to the teacher. Reading took 0.21 s per page. The reader is unaffected by rotation, perspective, compression, noise and shadows; it fails beyond 2 mm of warping, strong blur (out-of-focus photographs) and resolutions under 60 dpi, and does not detect ink lighter than 66% of the paper brightness, in which case the sheet is flagged because no document is found.

4. Impact

Daily practice. Aspa removes the manual steps between the question bank and the grades. Sheets are digitised with a departmental scanner or a phone, a class is read in seconds, and the teacher only handles the sheets flagged with a reason. Since grades are recomputed, voiding a flawed question or correcting a key costs one click and is documented in the exported workbook. The student review mode, with question texts and explanations, turns the exam review into a feedback session [6]. Local processing avoids the data-protection assessment of a third-party provider [9].

Table 3: Reading accuracy on synthetic sheets. Top: mixed defects. Bottom: largest level of each defect with all answers read exactly (30 sheets, 600 answers per level).

Mixed defects	Versions	Personalised
Sheets read (rest: cropped)	497/500	123/125
ID documents read	497/497	(printed)
QR codes read	496/500	123/125
Answers read exactly	9,939/9,940	2,126/2,126
Individual defect	**Largest level with 100% accuracy**	
Rotation	any angle	
Corner displacement / keystone	10% / 30% of page width	
JPEG quality / noise / shadow	5 / $\sigma = 40$ / 20% brightness	
Covered markers	1 of 4	
Resolution	120 dpi ($\geq$99.6% at 60 dpi)	
Paper warping	1.5 mm (98.3% at 2 mm)	
Blur (220 dpi)	$\sigma = 1$ px ($\geq$98% up to 2 px)	
Ink grey level (255 = white)	160 (undetected from 175)	

Research. Personalised exams make it feasible to apply, at scale and on paper, policies that exempt students from content already mastered [7], and to study their effect on quiz participation, grade distributions and the validity of the final exam. The exported per-option responses and the frozen versions support research on item quality and distractor analysis [2], order effects between versions and negative marking [3], since alternative scoring rules can be applied to the same readings.

Use. [Courses of the University of Cádiz in which Aspa has been used, number of exams and students graded, academic years, other adopters, download statistics and related publications or teaching-innovation projects.] Aspa is distributed under the AGPL licence; it is not used commercially.

5. Conclusions

Aspa covers the complete cycle of paper-based multiple-choice exams: versioned and personalised generation, optical reading of scans and phone photographs, grading with negative marking and partial credit, student review, item statistics with reversible adjustments and export. A shared sheet geometry and grades recomputed from plain files keep settings adjustable after the exam without losing corrections. Experiments show accurate reading under realistic defects, with residual errors flagged for review. Future work includes validation on large sets of real scans, support for more options and

questions per sheet, and export to learning management systems such as Moodle.

Acknowledgements

[Optional: colleagues and students who tested Aspa, and funding (e.g. teaching-innovation projects of the University of Cádiz).]

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Current executable software version

Aspa is run from source with the launch scripts of the repository; a standalone executable can be built with `herramientas/crear_ejecutable.py` [add the link to the release with executables, if published].